14. DIFFERENT LAYERS OF THE EYE 2

DURATION : 05:25

STUDY SHEET

COURSE SUMMARY

This video deeply explores the different layers of the eye, notably Cloquet's canal, the aqueous humour, the vitreous humour, and the development of the lens from the ectoderm. Key concepts include the importance of invagination in lens formation and the distinction between the anterior and posterior parts of the eye, comprising the lens vesicle and the retina. By revealing the connections between ocular structures, the video highlights complex embryological processes that enrich therapists' practice and their understanding of ocular pathologies.

THE DIFFERENT LAYERS OF THE EYE

There remains a **fine canal** in the eye, called the **Cloquet's canal**. This canal is a remnant in the vitreous humour.

The **aqueous humour** and the **vitreous humour** are two types of humours present in the eye. The vitreous humour can be imagined as a **gelatinous water balloon** that floats in the eye, with a fine canal at its centre that connects directly to the **lens**. This canal is also linked to the **optic nerve** or the **hyaloid canal**.

The lens originates from the surface **ectoderm**. During the fifth week of development, an **invagination** forms a **lens vesicle** surrounded by **vascularised mesenchyme**. The placode invaginates to become the lens, a process similar to that of **neural tube** formation. This movement leaves a space inside, which becomes the lens.

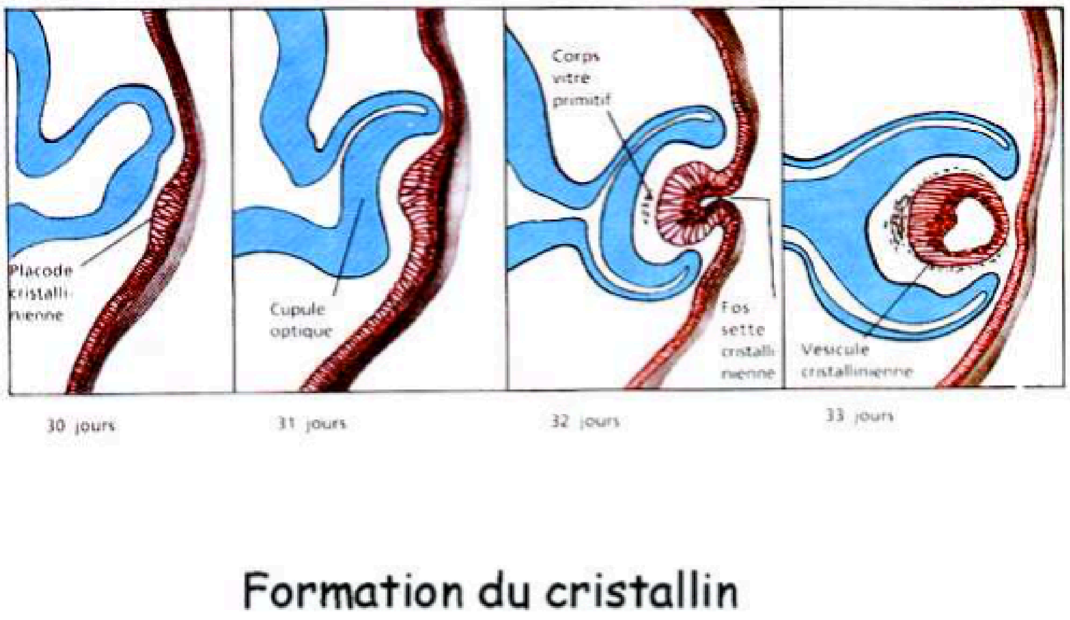

FIG. — Ectoderm invagination

The lens undergoes an evolution that forms **primitive lines**. It is essential to understand how it develops like a neural tube to become the lens of accommodation.

The **coloboma cleft** presents a line which, with the arteries, will give rise to Cloquet's canal. This canal closes with the **hyaloid artery** and **vein** which are located at the centre. The primary lens vesicle evolves to become a secondary lens vesicle with the placode.

On the outside, the layer formed is the **sclera**, which constitutes the first 'sock'. The **choroid** forms the second 'sock', connecting the lens. The eye can be divided into two parts: **anterior** and **posterior**. The anterior part is divided into the **anterior chamber** and the **posterior chamber**.

The **conjunctiva** and the **cornea** are continuous, with the cornea being transparent and continuous with the sclera. The choroid, also called **UV** (from the Latin *uva*, meaning grape), is a skin rich in blood vessels. It is divided into three parts: the choroid, the cornea at the front, and the **iris** at the back, which is a continuation of the choroid.

The **iris** is held in place by a **suspensory ligament** and **ciliary muscles**.

Finally, the **retina** is the innermost layer of the eye. It originates from the first vesicle, the **primary optic vesicle**, which develops by forming a cup-shaped structure. The retina is composed of two layers, inner and outer, and it is here that **retinal detachment** occurs.